

Implementation Report & Roadmap

WIS 2.0 Complementary Training Workshop - June 2026

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Executive Summary

This report presents the outcomes of the WIS 2.0 Complementary Training Workshop (June 2026), funded by CREWS and the Caribbean Meteorological Organization (CMO). It translates workshop findings into a concrete implementation roadmap for the Grenada Meteorological Service, including installation of a local WIS 2.0 node (wis2box) and a WIS2Downloader instance.

Critically, this roadmap has been designed to be realistic given our current operational reality: the office is operating with a reduced forecasting team, and no dedicated project time can be allocated at this stage. All timelines below reflect a phased, opportunistic approach - work progresses during available windows rather than requiring sustained dedicated blocks.

⚠ Staffing Constraint Acknowledgement: The meteorological office is currently short one forecaster. This roadmap deliberately avoids assuming any protected project time. Phases 1-2 can be executed opportunistically (~4-8 hours/month). Phases 3-4 will require management to formally allocate scheduled IT and Met staff blocks to proceed on time.

1. Workshop Overview

The WIS 2.0 Complementary Training Workshop served as a direct operational follow-up to foundational WIS 2.0 training. Funded by CREWS and hosted with CMO support, the workshop translated theory into hands-on regional capability. The four core modules covered during the workshop were:

- **Module 1 - Ingestion & Node Management**
 - Hands-on wis2box configuration on European Weather Cloud virtual environments
 - Data workflow management, automated conversion, and monitoring pipeline setup
- **Module 2 - Pipeline Automation & Data Types**
 - Regional case studies: St. Maarten, Montserrat, Belize
 - Python scripting and SURFACE CDMS for automated pipelines

- Data types: DAYCLI, CLIMAT, upper-air, aviation, restricted access data
- **Module 3 - Data Retrieval & Forecast Integration (WIS2Downloader)**
 - Installation and deployment of WIS2Downloader
 - Custom subscriptions, filtering, access controls
 - Decoding WMO binary formats and integrating feeds into operational forecasting
- **Module 4 - Transition, Security & Sustainability**
 - GTS header mapping to WIS 2.0 notifications
 - Long-term maintenance protocols, software updates, data security
 - CMO Community of Practice alignment and post-training milestones

2. Strategic Context & Rationale

2.1 Why a Local Node Matters

The CMO currently operates a regional WIS 2.0 node on behalf of member states. While this provides an immediate pathway to compliance, it represents a single point of failure for Grenada's international data obligations.

The CMO regional node is hosted on Amazon Web Services (AWS). A catastrophic hurricane or major cloud outage could sever Grenada's ability to transmit observational data to the global network at precisely the moment it is most needed, during a severe weather event.

Establishing a local wis2box node addresses this directly by providing:

- An independent, locally-controlled publication pathway for all observational data
- Resilience against regional cloud infrastructure failures
- Full national sovereignty over data timing, licensing, and access controls
- A platform to meet CMO-mandated targets for DayCLI and CLIMAT transmission

2.2 The GTS Decommissioning Deadline

The Global Telecommunication System (GTS) - the current backbone of international meteorological data exchange is scheduled for decommissioning by 2033. WIS 2.0 will replace it as the sole global exchange mechanism. This transition directly affects the data feeds that populate our forecasting workstations (IBM, METLAB, SMARTMET) and various stakeholders that utilize our observation and forecast data.

Data currently accessed through third-party aggregator websites (e.g., ogimet for SYNOP/METAR) will need to be sourced directly through WIS 2.0 subscriptions. Beginning this transition now provides several years to build capability before the hard deadline.

2.3 WIS 2.0 Data Categories

Data on the WIS 2.0 network is organized into two streams with different publication and licensing requirements:

Data Type	Format/Method	Access	Target Phase	Status
Hourly Observations & DayCLI	CSV → BUFR conversion	Free / Global Cache	Phase 3	Core
CLIMAT & Upper-Air	BUFR format	Free / Global Cache	Phase 5	Core
CAP Alerts	XML push	Free / Global Cache	Phase 4	Core
Satellite Data Products	WIS 2.0 subscription	Free / Global Cache	Phase 2	Core
METAR & TAF (Aviation)	IWXXM format	Restricted - License Required	Phase 5	Recommended
NWP (GRIB/GRIB2, NetCDF)	WIS2Downloader subscription	Free / Global Cache	Phase 2	Core

3. Local Operational Priorities

3.1 Carriacou Airport — Hybrid Station Integration

Transmitting SYNOP data from Carriacou Airport via WIS 2.0 requires a unique WIGOS station identifier to be assigned to that location. Additionally, because the airport operates only during daylight hours, a hybrid station configuration is recommended to be implemented (If not, the metadata must clearly identify it is not 24 hours):

- Daytime: Human observer data published as SYNOP
- Overnight (automatic cutover): AWS data published continuously until the morning shift resumes
- Recommended solution: Leverage existing SURFACE CDMS functionality to manage this cutover automatically.

3.2 AWS Expansion & GOES Satellite Telemetry

To eliminate the recurring credential and access issues associated with our current NOAA DADS account, some workshop participants recommend transitioning to a direct satellite data pipeline:

- Request a NOAA LRGS (Local Readout Ground Station) account for direct GOES satellite data streaming
- Establish structured local storage with automated retention and quality control cycles
- Deploy SURFACE CDMS for data visualization, transmission, and station health monitoring
- Register all new stations on OSCAR and obtain WIGOS IDs
- Use WIS 2.0 subscriptions to pull NWP data (GRIB/GRIB2, NetCDF) directly to forecasters.

3.3 CDMS Modernization - Four-Stage Architecture

Our current legacy web application for AWS data viewing is operationally inadequate. The workshop outlined a modern four-stage data pipeline architecture that all NMHSs should work towards:

STAGE 1	STAGE 2	STAGE 3	STAGE 4
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AWS Data Loggers Address manufacturer inconsistencies in sample rates, QC checks, and storage intervals.	Reception & Storage Evaluate data-transfer protocols and establish structured local storage with automated retention.	CDMS - Single Source of Truth WMO No. 1131-compliant CDMS. Automated QC, and push of DayCLI/CLIMAT/Hourly to wis2box.	WIS 2.0 & Stakeholder Access Modern APIs for data sharing, formal local data policies, and WIS 2.0 publication.
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4. Implementation Roadmap

The following phased roadmap reflects the current operational reality of the Grenada Meteorological Service. With staffing running below capacity, phases are designed to be executed opportunistically progressing during available windows between operational duties rather than requiring dedicated protected time, particularly in the early stages.

Phases 1 and 2 are deliberately low-intensity and low-risk. They can be advanced by any staff member with basic IT access during quiet operational periods. Phases 3 and 4 will require formal management commitment of scheduled time blocks to complete within the target timeframe.

Phase	Timeframe	Focus	Key Tasks	Est. Effort	Owner
Phase 1 LOW RISK	Months 1-2	Foundation & Planning	- Hardware/VM procurement request - IT briefing & firewall planning - WSL 2 + Docker setup on available workstation - WIS2Downloader test install (low-risk, read-only)	~4-6 hrs/month (<i>opportunistic</i>)	Met Staff + IT
Phase 2 MEDIUM	Months 3-5	WIS2Downloader Deployment	- Deploy WIS2Downloader on dedicated machine - Configure data subscriptions (NWP, SYNOP, satellite) - Integrate data feeds into forecasting workstations - Validate incoming data quality	~6-8 hrs/month (<i>opportunistic</i>)	Met Staff + IT
Phase 3 MEDIUM	Months 6-9	wis2box Node Installation	- Install & configure wis2box (Linux/WSL2 environment) - Publish metadata records to global discovery - Configure SYNOP/AWS automated pipelines - Test BUFR conversion for hourly obs & DayCLI	~8-12 hrs/month (<i>scheduled blocks</i>)	Met Staff + IT
Phase 4 HIGH FOCUS	Months 10-12	Validation & Go-Live	- End-to-end testing of data publication - CAP message integration - Carriacou hybrid-station cutover logic - Staff monitoring training	~10 hrs/month (<i>scheduled blocks</i>)	Met Staff + IT

Phase	Timeframe	Focus	Key Tasks	Est. Effort	Owner
			– Document operational procedures		
Phase 5 FUTURE	Year 2+	Expansion (Post-stabilization)	– CLIMAT & upper-air BUFR publishing – IWXXM aviation data (METAR/TAF) – SURFACE CDMS integration – NOAA LRGS account & GOES telemetry pipeline	Ongoing (dedicated time needed)	Met Staff + IT + Management

Key Assumptions & Dependencies

- A dedicated Linux-capable machine or server (physical or virtual) is procured or allocated before Phase 3 begins. wis2box cannot run on a standard Windows workstation without WSL 2 (untested) or a VM/Linux workstation, but is recommended to be placed on a server.
- IT department engagement is secured before Phase 2. Firewall configuration and network security cannot be handled by Met staff alone.
- Management formally endorses this roadmap and acknowledges that Phases 3-4 require scheduled, protected working time to complete on target.
- CMO and COWET peer support networks remain accessible for remote troubleshooting (a key benefit of the Community of Practice established at this workshop).

5. Capacity Building & Security Requirements

5.1 IT & Personnel Requirements

WIS 2.0 is not a one-time installation, it is a continuously maintained operational system. Sustainable implementation requires:

- Dedicated IT involvement (or dedicated IT personnel for the department), formal allocation of protected IT hours, covering: Linux environment management, Python scripting, database administration, network and firewall configuration.
- Met staff must be trained in maintaining ongoing familiarity with wis2box operations, monitoring dashboards, and BUFR conversion pipelines
- Management recognition that this is infrastructure investment, not a short-term project

5.2 Network Security

The WIS2Downloader, once deployed, will maintain persistent outbound connections to global WIS 2.0 brokers. Proper security configuration is essential:

- IT must open only the specific ports required by WIS 2.0, no broad firewall exceptions.
- Stringent access controls must be applied to local servers running wis2box and WIS2Downloader to prevent external traffic overload or unauthorized access

- Regular monitoring of system logs and data throughput must be part of the standard operational routine

5.3 Compliance Monitoring & GBON Obligations

Our standing within international support programs, particularly the Systematic Observations Financing Facility (SOFF), is directly tied to demonstrable, continuous compliance with Global Basic Observing Network (GBON) data publication requirements. Staff must:

- Monitor real-time dataflows through the wis2box dashboard
- Identify and respond to data-flow gaps before they result in compliance breaches
- Maintain records of publication metrics for CMO and WMO reporting

6. CMO Strategic Recommendations

The following recommendations were formally agreed at the regional level and directly inform Grenada's implementation priorities:

Recommendation 1 - National Milestones

- National WIS 2.0 Node deployment
- National metadata update and publication
- Increased operational data exchange
- CAP alert dissemination via WIS 2.0
- Data consumption subscriptions

Recommendation 2 - Regional Publication Targets

Phase 1 priorities (immediate focus):

- SYNOP and AWS observations on national nodes
- Upper-air observations
- CAP messages
- DAYCLI
- CLIMAT

Phase 2 priorities (medium-term):

- IWXXM aviation products dissemination
- Radar products dissemination
- NWP products

Recommendation 3 - IWXXM Aviation Migration

CMO is co-developing a TAC-to-IWXXM Translator. Grenada should support the progressive migration of METAR/TAF to IWXXM format via WIS 2.0, participating in pilot implementations as capacity allows (targeted for Phase 5 of this roadmap).

Recommendation 4 - Regional GitHub Repository

COWET will establish a version-controlled regional knowledge and code repository on GitHub for sharing scripts, automation workflows, technical documentation, and lessons learned. Grenada's technical staff should actively contribute and draw from this resource throughout implementation.

Recommendation 5 - Annual In-Person Workshops

COWET will convene at least one in-person workshop annually over the next three years, supplemented by at minimum, three virtual meetings per year. These provide hands-on troubleshooting opportunities and peer-to-peer learning that are essential for sustained implementation progress.

Recommendation 6 - Regional ICT Support Facility

CREWS Caribbean 3.0 should establish a dedicated Regional ICT/WIS 2.0 Specialist to provide on-demand technical support, mentoring, and implementation assistance to member states. This is particularly relevant for Grenada given current staffing constraints.

7. Recommendations to Management

The items below represent the minimum management decisions required to enable this roadmap to proceed. Technical staff cannot advance beyond Phase 2 without these commitments.

#	Recommendation	Priority / Timing
1	Formally endorse this implementation roadmap and communicate timelines to all relevant staff and IT and consult with regional team so that IT department can have a better understanding of the requirements.	Immediate
2	Procure or designate a dedicated server or virtual machine (Linux-capable) for wis2box and WIS2Downloader deployment before Phase 3.	Before Month 6
3	Formally engage the IT department and allocate protected hours for WIS 2.0 security and infrastructure work.	Before Month 3
4	Initiate the WIGOS ID registration process for Carriacou Airport to enable SYNOP data publication.	Phase 1
5	Initiate a NOAA LRGS account request to transition away from the DADS dependency for AWS data ingestion.	Phase 1-2
6	Advocate with CMO for Grenada to be an early beneficiary of the Regional ICT Support Facility when established.	Ongoing

8. Conclusion

The WIS 2.0 Complementary Training Workshop has provided the Grenada Meteorological Service with both the technical knowledge and a clear regional mandate to begin local WIS 2.0 implementation. The transition away from GTS is not optional, it is a WMO-driven global requirement with a firm 2033 deadline.

This report recommends a phased, staffing-aware roadmap that begins immediately with low-effort, high-value steps (WIS2Downloader deployment and hardware planning) and progressively builds toward a fully operational local wis2box node. The approach is designed to make meaningful progress without placing unsustainable demands on a team already operating below capacity.

The investments required in hardware, IT time, and management commitment are modest relative to the operational and compliance risks of inaction. Grenada's data sovereignty, forecasting capability, and standing within international support programs all depend on completing this transition.

The next concrete action is management endorsement of this roadmap, followed by an IT briefing to initiate firewall planning and hardware assessment. Technical staff are ready to begin Phase 1 tasks as soon as those decisions are made.